IA2 - AP_DS - Tutorial 03- Python Programming Essentials

Project 01

Question 01 : Clean Lines Function

File to create: utils/helpers.py

Task:

Write a function called clean_lines() that:

- Takes a list of strings
- Removes empty lines and trims extra spaces
- Returns the cleaned list
- Give a solution first without using a list comprehension, then using a list comprehension.

Example Input:

lines = ["Laptop,1000,3", " ", "Phone,500,5", "\n", "Tablet,700,2"]

Expected Output:

["Laptop,1000,3", "Phone,500,5", "Tablet,700,2"]

Question 02: Parse Products Function

File: utils/helpers.py (add below previous function)

Task:

Write a function parse_products(lines) that:

- Converts each line into a tuple (name, price, quantity)
- Converts price to float and quantity to int
- Ignores malformed lines

Example Input:

lines = ["Laptop,1000,3", "Phone,500,5", "Tablet,700,2"]

Expected Output:

[("Laptop", 1000.0, 3), ("Phone", 500.0, 5), ("Tablet", 700.0, 2)]

Question 03: Product Class

File to create: utils/product.py

Task:

Create a class Product with:

- Attributes: name, price, quantity
- Method: total_value() that returns price * quantity

Example Usage:

p = Product("Laptop", 1000, 3) print(p.total_value()) # Output: 3000

Question 04: Create Products from File

File: scripts/create_products.py

Task:

Read a text file data/raw/data.txt containing product info (name,price,quantity)

- Clean the lines using clean_lines()
- Parse the products using parse_products()
- Create Product objects
- Print each product's total value and overall total sales

Example Input File (data/raw/data.txt):

Laptop,1000,3

Phone,500,5

Tablet,700,2

Monitor,300,4

Camera,400,1

- **Expected Output**:

Laptop: 3000.0

Phone: 2500.0

Tablet: 1400.0

Monitor: 1200.0

Camera: 400.0

📊 Total Sales: 8500.0

Question 05: Notebook for Visualization

File: notebooks/analysis.ipynb

Task:

- Read data/raw/data.txt
- Create Product objects
- Plot a bar chart of total sales per product

Plot: X-axis: product names Y-axis: total sales

✅ Folder Structure for Exercises

```
Sales_project/
│
├── data/
│   ├── raw/
│   │   └── data.txt
│   └── processed/           # optional for cleaned files
│
├── scripts/
│   └── create_products.py
│
├── utils/
│   ├── __init__.py
│   ├── helpers.py
│   └── product.py
│
└── notebooks/
    └── analysis.ipynb ```
```

Project 2 (Homework)

Create a small Python project to simulate and analyze data for **10 students**. For each student, generate:

- a student ID;
- a random birth date between 2000 and 2006;
- three random grades between 0 and 20;
- the student's age;
- the average of the three grades.

Then perform a statistical analysis of the class, including:

- mean;
- median;
- standard deviation;
- minimum and maximum average;
- mean age.

Finally, save a summary of the results in a text file.

Project Structure

```
student_project/
│
├── main.ipynb
├── students.py
├── analysis.py
└── results/
    └── report.txt
```

Requirements

- `students.py` must contain the functions related to generating student data.
- `analysis.py` must contain the functions related to calculating ages and statistics.
- `main.ipynb` must import and use these functions, display the generated data, and present the results.
- The program must automatically create the `results` folder if it does not exist.
- The final analysis must be saved in `results/report.txt`.